

attempting to maximise data centre efficiency.

Do the same rules apply to industrial and automotive sectors as well?

Shashwath: Our thesis underscores the notion that data centres and mobile devices have long dominated the tech landscape, inadvertently neglecting critical features in other sectors like industrial and automotive. These relatively neglected aspects include functional safety and security, paramount in industries beyond the relentless pursuit of speed and power. The potential for significant financial gains in these sectors also drives the pursuit of speed. When the risk-to-reward ratio is favourable, prioritising those areas makes sense. Over the past 15 years, the focus has been on optimising for smartphones, but the landscape is evolving. There is no longer a need for universal smartphone optimisation. Instead, the emphasis should be on feature optimisation and tailoring power, performance, and area considerations to the specific use case. As long as it meets the requirements of the given application, it is deemed sufficient.

What are the factors to consider while designing a chip for the automotive sector?

Sharan: Automotive regulations and design requirements are among the most stringent, only exceeded by those in military and space applications. While the military domain often operates at lower volumes, which sets it apart, the automotive industry requires top-tier quality and compliance at a significant scale. Automotive companies typically commit to a chip or a specific hardware configuration for a decade or more. They require long-term support, extensive testing, and other assurances. Our first chip is not intended to be automotive-certified. However,

automotive certification is a significant consideration in our approach to design. We aim to align ourselves with this high-quality standard, although achieving it at scale may take time. While 60% of automotive quality falls short of full automotive-grade certification, it represents a high level of quality, especially for microcontroller-class chips. Offering this level of quality within the microcontroller price range means that a chip can withstand heat and humidity, maintain speed, and remain stable, making it a robust and dependable choice for various use cases involving microcontrollers.

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What are the factors to keep in mind to maintain power efficiency and a low-power design?

Sharan: Efficiency can be prioritised by avoiding unnecessary components and features in the chip design. At Mindgrove, we opt for the 28-nanometre process for microcontrollers rather than the 65-nanometre or 45-nanometre options that are commonly used. This unconventional choice challenges us to create a well-designed chip capable of delivering the required performance without resorting

to excessive complexity or gate counts, resulting in power efficiency. Our primary emphasis is on optimising performance, and then we progressively eliminate components and features until we achieve the desired power consumption. We aim to create a chip perfectly suited for its intended purpose, effectively a 'right-sized' chip for the task at hand.

How would domestic semiconductor manufacturing impact chip design and development?

Shashwath: If a foundry capable of meeting our requirements were to emerge in India, it would be highly appealing. This could significantly reduce costs and offer advantages in terms of both time and the logistics supply chain compared to dealing with overseas foundries. International dealings can be cumbersome. It is important to minimise lead times in the supply chain to ensure swift product delivery, especially when dealing with large-scale operations involving millions of shipments annually. This can lead to substantial cost savings that significantly affect the bottom line. Another intriguing prospect is if someone sets up ATMP (assembly, test, and packaging) facilities in India. We do have such facilities but they are just focusing on entry-level capabilities, which are relatively less challenging to set up.

Sharan: Starting ATMP facilities with QFN or QFP packaging can be a significant step. These package types already have a substantial market in India. However, it becomes even more compelling when it comes to BGA (ball grid array) packaging. This aligns with the considerations of power, performance, and area calculations. BGA packages are particularly advantageous when transmitting data rapidly and efficiently. **EFY**